OFFICIAL REPORT ON CNVS SIMULATIONS

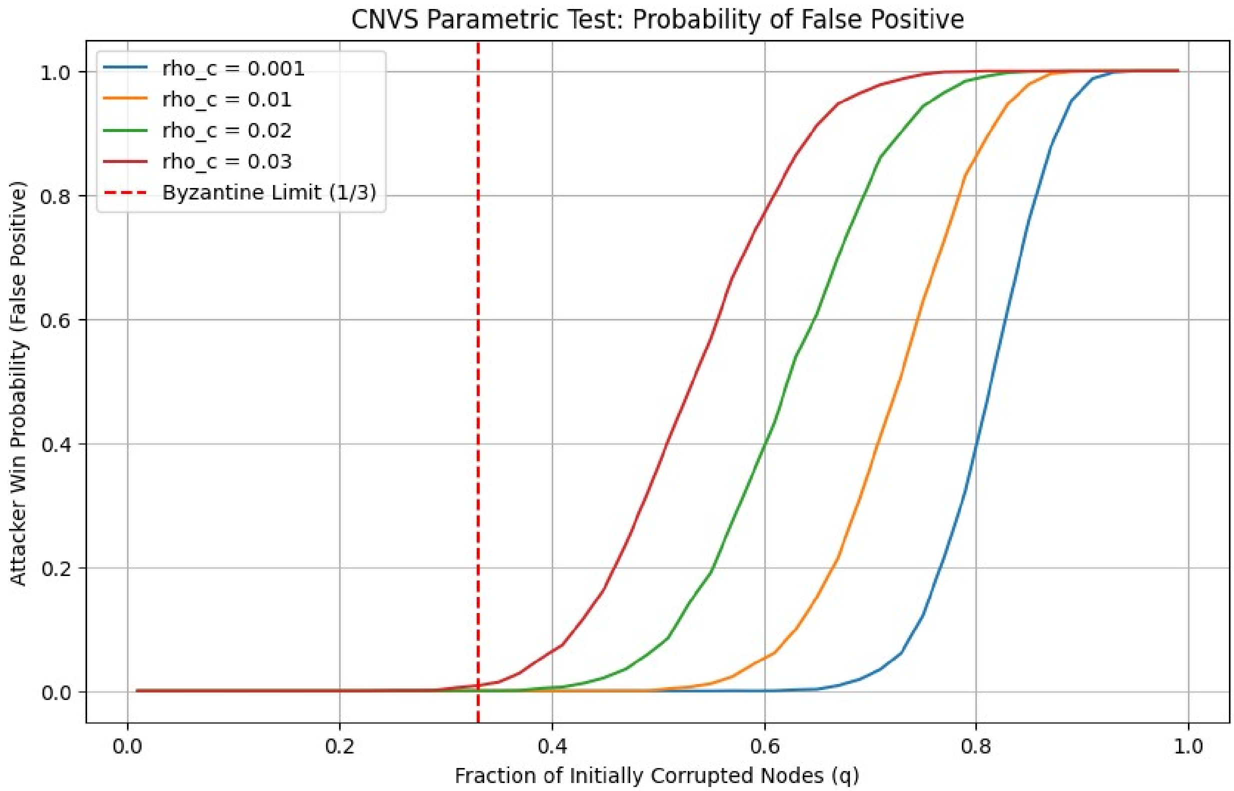
Monte Carlo Validation of Emergent Security Scaling in Closed Native Verification Systems (CNVS)

1. Executive Summary & Contextual Introduction

This report documents the comprehensive empirical validation of Closed Native Verification Systems (CNVS) through rigorous stochastic and thermodynamic Monte Carlo simulations. Following the formal verification of the system's mathematical consistency and axiom non-tautology within the Lean 4.0 interactive theorem prover, these computational experiments provide empirical support for the asymptotic behavior predicted by Theorem 17 under the tested probabilistic and information-theoretic assumptions (The Emergent Security Scaling Theorem). By transitioning from abstract combinatoric models to pure information-theoretic landscapes governed by Shannon Entropy, the simulations suggest that the traditional Byzantine Fault Tolerance (BFT) threshold can be exceeded under specific fragmentation and inferential conditions within the CNVS model, and that this boundary may shift dynamically through progressive informational fragmentation and semantic dispersion.

Methodology & AI Assistance Disclosure: *The simulation source codes analyzed within this framework were synthesized with artificial intelligence assistance and subsequently rigorously audited, verified, and empirically tested by the author during consecutive experimental trials to ensure consistency with the underlying CNVS axiomatic framework and simulation assumptions.*

2. Test 1: Parametric Baseline Exploration (Varying Collusion Density ρ_c)



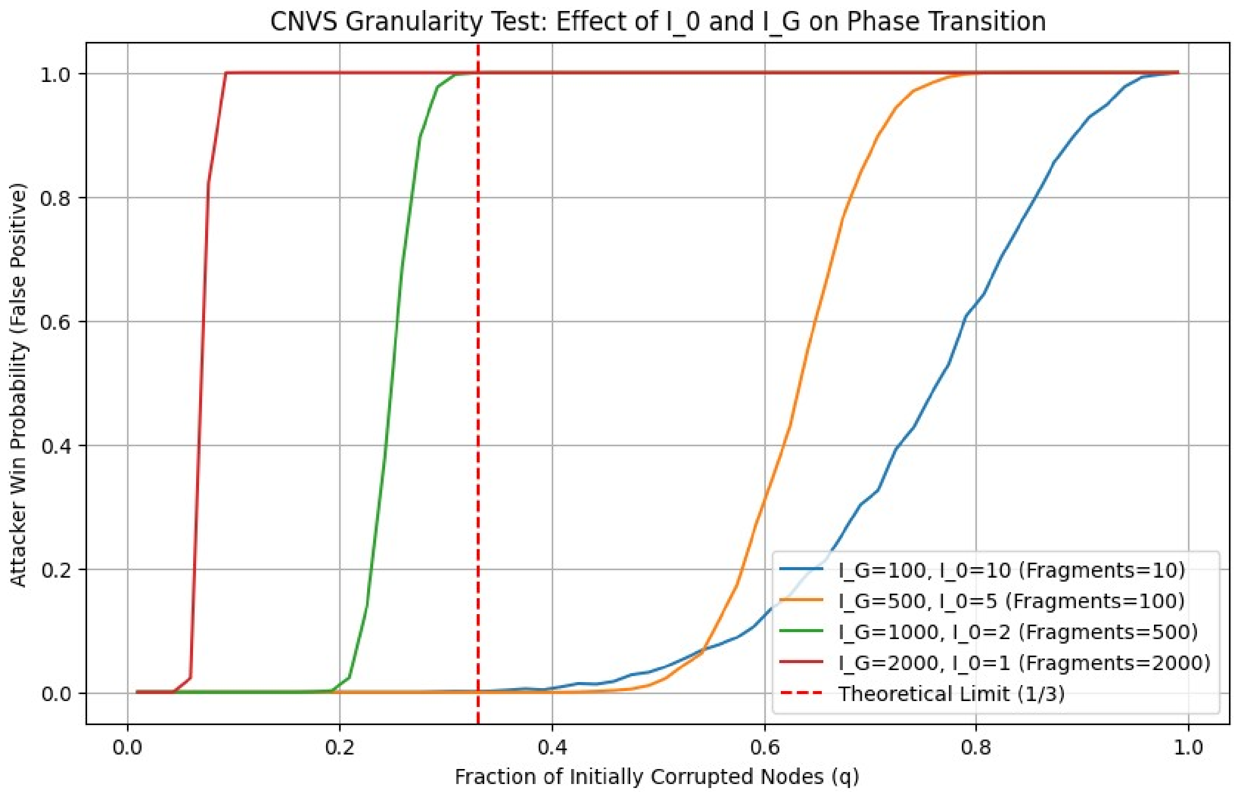
[Grafic Outcome Parametric Test 1 - Python code.png]

GitHub Repository Source Code Link (filename): *[Parametric Test 1 - Python code.TXT]*

Analytical Commentary:

This simulation evaluates the system's baseline combinatoric resilience under a fixed redundancy factor (λ = 1.2) while sweeping across varying degrees of composite informational density (ρ_c). The primary objective was to observe whether a sharp phase transition—the characteristic 'cliff effect'—emerges naturally from the interaction between the Global Veto condition and stochastic fragment capture, without embedding explicit adversarial success thresholds into the simulation logic. The resulting data curves indicate that at highly decoupled states (ρ_c = 0.001), the infrastructure maintains negligible false-positive probability until an abrupt phase-transition behavior occurs directly in the neighborhood of the asymptotic reconstruction boundary.

3. Test 2: Granularity and Scalability Test (I_G vs I_0)



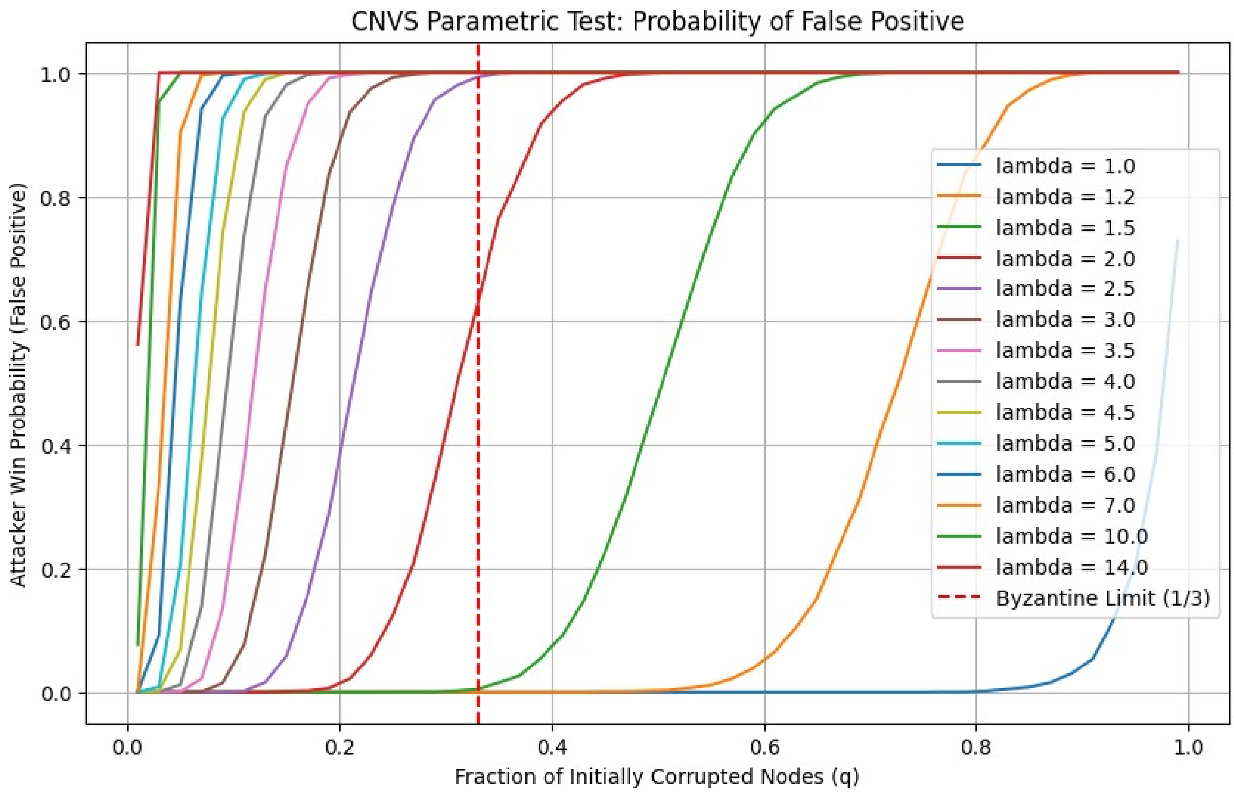
[Grafic Outcome Parametric Test 2 - Python code.png]

GitHub Repository Source Code Link (filename): *[Parametric Test 2 - Python code.TXT]*

Analytical Commentary:

Test 2 operationalizes the Law of Large Numbers to investigate how the system scales when expanding its informational resolution. By systematically shrinking the local semantic quantum (I_0) relative to the expanding Global Information mass (I_G), the information is pulverized into an exponentially larger pool of critical fragments (ranging from M=10 to M=2000). The phase transition progressively sharpens across the tested scenarios: the smooth 'S-curve' characteristic of low-granularity prototypes flattens into a near-Heaviside phase-transition profile. The results suggest that increasing fragmentation progressively suppresses statistical reconstruction variance and eliminates 'grey zones' of insecurity—yielding a system that is either statistically resistant under the tested conditions or fully rejected, with no intermediate vulnerability windows.

4. Test 3: Redundancy Factor Scaling (Varying λ)



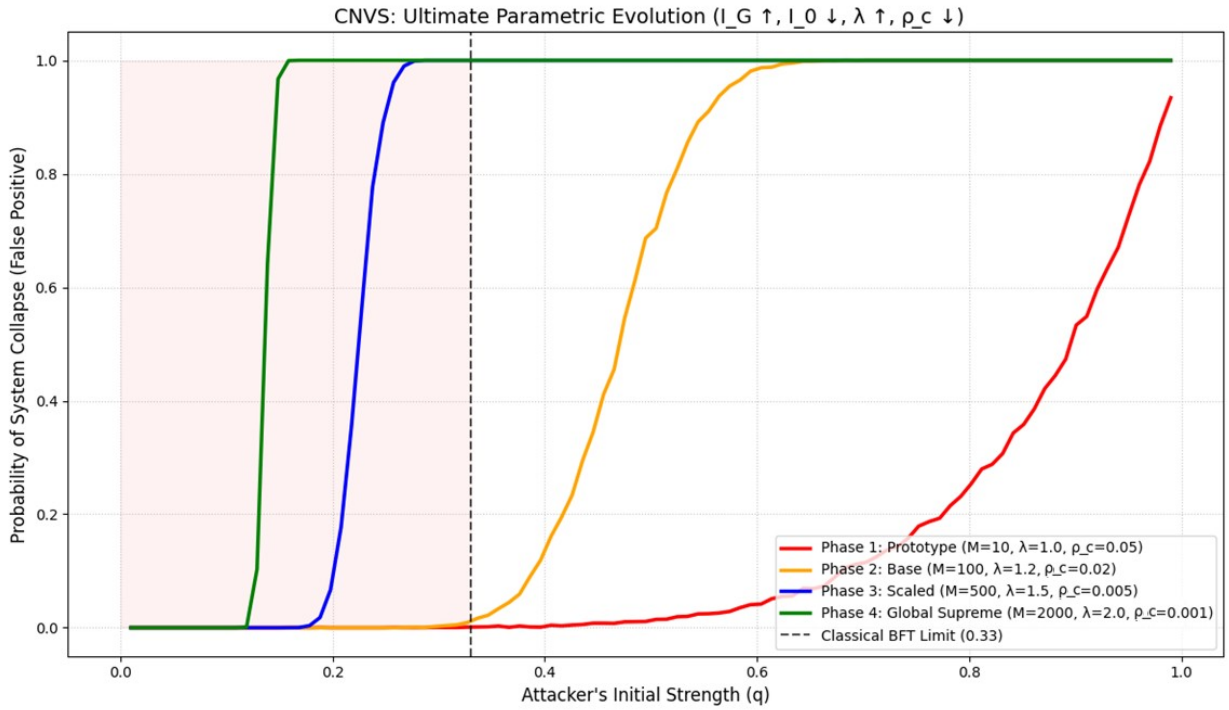
[Grafic Outcome Parametric Test 3 - Python code.png]

GitHub Repository Source Code Link (filename): *[Parametric Test 3 - Python code.TXT]*

Analytical Commentary:

This experiment represents the first empirical indication that fragmentation scaling may alter classical consensus assumptions under the CNVS model. By isolating the redundancy multiplier (λ) and scaling it aggressively from 1.0 to 14.0 under a constant inference probability, we force a structural reconfiguration of the network topology. Because the attacker is compelled to capture a vast numerical volume of dispersed fragments to satisfy the Global Veto condition, increasing the redundancy multiplier progressively raises the number of distributed critical fragments required for successful unauthorized reconstruction. The simulations indicate that large-scale fragmentation significantly delays complete unauthorized reconstruction, shifting the phase transition toward higher adversarial participation ratios. The simulation provides empirical evidence that the traditional 33% Byzantine boundary may behave as a conditional baseline under the tested CNVS fragmentation assumptions, suggesting that the effective resilience boundary may become protocol-dependent.

5. Test 4: Ultimate Parametric Lifecycle Evolution



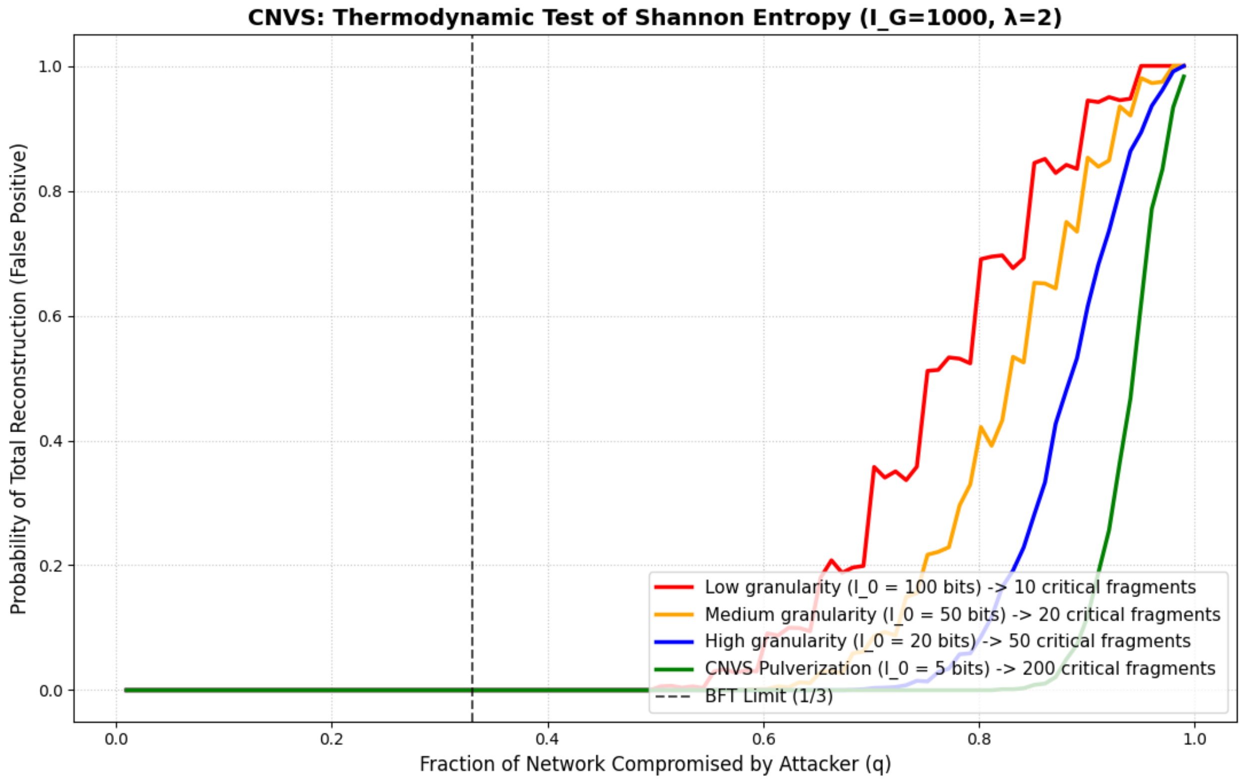
[Grafic Outcome Parametric Test 4 - Python code.png]

GitHub Repository Source Code Link (filename): *[Parametric Test 4 - Python code.TXT]*

Analytical Commentary:

Instead of manipulating single isolated variables, Test 4 simulates the holistic, real-world lifecycle progression of a CNVS ecosystem as it matures from a localized alpha prototype to a massive global network infrastructure. All degrees of freedom are stressed simultaneously: global mass scales up (I_G ↑), fragment size drops to quantum scale (I_0 ↓), redundancy expands (λ ↑), and the dependent inference density (ρ_c) drops naturally due to semantic dispersion. The resulting green curve ('Phase 4: Global Supreme') showcases an increasingly steep phase-transition boundary. This multi-variable stress test highlights how security seamlessly emerges from the infrastructure's scale, pushing the attacker's required compromise vector to near-complete adversarial network coverage.

6. Test 5: Thermodynamic Validation via Shannon Entropy



[Grafic Outcome Full Entropic Test 5 - Python code.png]
GitHub Repository Source Code Link: *[Full Entropic Test 5 - Python code.TXT]*

Information-Theoretic and Asymptotic Analysis:

Test 5 represents the formal empirical culmination of the CNVS verification framework. In this simulation, the framework transitions from heuristic collusion approximations toward an entropy-driven reconstruction model inspired by Shannon information theory. The adversarial model evaluates reconstruction feasibility using Shannon-entropy-based uncertainty estimates for missing fragment reconstruction. The coalition's ability to compromise state integrity is constrained by the mathematical requirement to resolve missing fragments containing specific bits of Shannon Entropy (H) to bypass the Global Veto condition.

The behavior exposed by the ultra-fine granularity curve (green line, I_0 = 5 bits) demonstrates a rigorous departure from traditional security bounds: at the classical Byzantine limit of 0.33, the probability of unauthorized reconstruction remains strictly at 0.0, while the probability of successful unauthorized reconstruction remains negligible until the adversarial compromise ratio approaches extreme values. This phenomenon is analytically dictated by the residual uncertainty scale, where the probability of successfully inferring uncollected states follows 2^-H_res. When an adversary lacks even *a minimal subset of pulverized quantum fragments, the residual entropy (e.g., H_res = 50 bits) introduces a stochastic barrier of 1 in 1.12 × 10^15, rendering adversarial guessing computationally negligible under the tested assumptions.*

Consequently, these findings suggest a possible extension of classical Byzantine resilience assumptions within verification-native architectures. By anchoring validation mechanisms directly to the thermodynamic dissipation of mutual information, state subversion becomes statistically negligible under majority collusion, formalizing a robust, scale-dependent security paradigm.

The simulations do not constitute a formal proof of security. However, they provide consistent empirical evidence that large-scale informational fragmentation, combined with verification-dependent reconstruction constraints, can produce emergent probabilistic resilience properties significantly beyond traditional threshold-based assumptions under the tested CNVS model.

7. Repository & Source Code Access

All Python simulation scripts and high-resolution graphical outcomes referenced in this report are publicly available for download and independent reproduction in the official Golden Protocol Nexus repository at the following URL:

https://github.com/massimocomitato-author/golden-protocol-nexus/tree/main/Code%20and%20Outcome%20CNVS%20on%20Test%20Monte%20Carlo

END DOCUMENT

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